

High-prime isogeny continuation

P-192 / P-224 | 2026-10-09 | independently replayed maps

Result: P-192 degree 10453 and P-224 degree 1471. Each is an explicit rational map over its source prime field, verified independently by the existing walker arithmetic. One of the two eigenlines is constructed at each degree; full enumeration remains unresolved.

P-192: kernel degree 5226; numerator 10454 coefficients, denominator 10453 coefficients. Kernel, codomain, exact rational-map substitution and 20 fresh public scalar transports pass; the mapped nonidentity generator satisfies the published prime subgroup order.

P-224: kernel degree 735; numerator 1472 coefficients, denominator 1471 coefficients. Kernel, codomain, exact rational-map substitution and 20 fresh public scalar transports pass; the mapped nonidentity generator satisfies the published prime subgroup order.

Method: choose a recorded low-order Frobenius eigenline, construct prime-degree torsion over a validated degree-three or degree-five finite extension, verify its order/eigenvalue, descend its cyclic kernel to the base field, and use the native Kohel map API. All extension and map checks remain explicit.

Preserved full-Hecke attempts: P-192 degrees 509 and 1021, P-224 degrees 521 and 1031. They timed out under the preregistered 600- or 1800-second budgets. They still required two maps.

Kernel-first records separately require one map and retain PARTIAL coverage.

Bounds: one construction at a time; seed 1; sampled RSS cap 8 GiB, 25 ms samples, at most 100 ms transient monitor-unavailability grace. Kernel cases each had 1800 seconds. Elapsed receipts are operational records on a shared host, without a matched speed or ECDLP-cost comparison.

Evidence: KERNEL_PROTOCOL.md, PROTOCOL.md, COMMANDS.md, kernel-p192-10453/replay.json, kernel-p224-1471/replay.json and each run's raw CLI record and receipt. The prior 2026-10-08 report and all failures remain preserved.

Verified prime-degree isogenies beyond 1009

One independently verified Frobenius eigenline per source; second eigenline unresolved.

P-192: Source

icv1-fp192-t31607402316713927207482677199-52e4af59

EC1P192Cp192h5531c4a08bdb

urn:ec-record:1:sha256:5531c4a08bdb64b6e86a6e30e9a08a
a57edef7af15ac5f6d4d2a83a53bf2f646

Kernel, exact map and subgroup transport: PASS

degree 10453 ->

P-192: Verified target

icv1-fp192-t31607402316713927207482677199-e2f29854

EC1P192Cfph565b24cb61ba

urn:ec-record:1:sha256:565b24cb61baf7e082d65cad3c41cd
468ec662b8bfe6a0e76d27f66925d4caea

Kernel, exact map and subgroup transport: PASS

P-224: Source

icv1-fp224-t4733100108545601916421827343930821-fec01e99

EC1P224Cp224h8e110c585480

urn:ec-record:1:sha256:8e110c5854808bfd2283aa4b15063d
4c3052a0d742a47224e34876c6ade763bd

Kernel, exact map and subgroup transport: PASS

degree 1471 ->

P-224: Verified target

icv1-fp224-t4733100108545601916421827343930821-5a35f6f7

EC1P224Cfph266bc36adb82

urn:ec-record:1:sha256:266bc36adb82a7afb36a60c7b868ce
b1f9d18d757bdb8c6b8e4b102bd403632d

Kernel, exact map and subgroup transport: PASS

Completed explicit probes (logarithmic prime-degree axis)

Population: six preregistered construction attempts. Green: verified map. Orange: timeout.



Verified individual maps retain PARTIAL degree coverage; the other eigenline is not drawn.

Validation and provenance

114 standalone release tests pass. The supervisor's two structural/order tests and all four extension tests pass. Reducible moduli are rejected; source-sized extension inverses and Frobenius relations pass; an exact small-field point census independently checks the trace recurrence.

The field-valued divisor sieve removes redundant divisor scans and u64 overflow from modular-series setup. Optional stderr stage traces preserve the later construction bottlenecks. This stage correction is not reported as an end-to-end speed measurement.

The independent verifier retains squarefreeness, division-polynomial torsion, subgroup closure, the Velu codomain, normalization/coprimalty, exact curve-equation substitution and fresh public subgroup transport. Exact Karatsuba, reciprocal reduction, lazy recurrences and block composition agree with the unchanged reference routines; 12 verifier tests pass.

P-224's certificate completed on the original all-index recurrence. The initial eager P-192 replay was stopped before producing a certificate, and its raw construction and logs remain preserved. The memoized exact verifier then completed all checks for degree 10453.

Existing registry records and IC measurements are preserved. New target models are registered with their mapped generators and exact EC1/UIDs, then native covers, aliases, leaderboard rosters and browser identities are refreshed. Root-library and dependent traits export gates are recorded honestly in EXECUTION_NOTES.md.

Sources: the native algorithm API and prior walker modules, KERNEL_PROTOCOL.md and the committed source snapshots. report.rs is the editable report/vector/PDF source. MANIFEST.json binds source, executables, corpus bytes and canonical outputs. PDF pages are rendered with Poppler and visually checked.